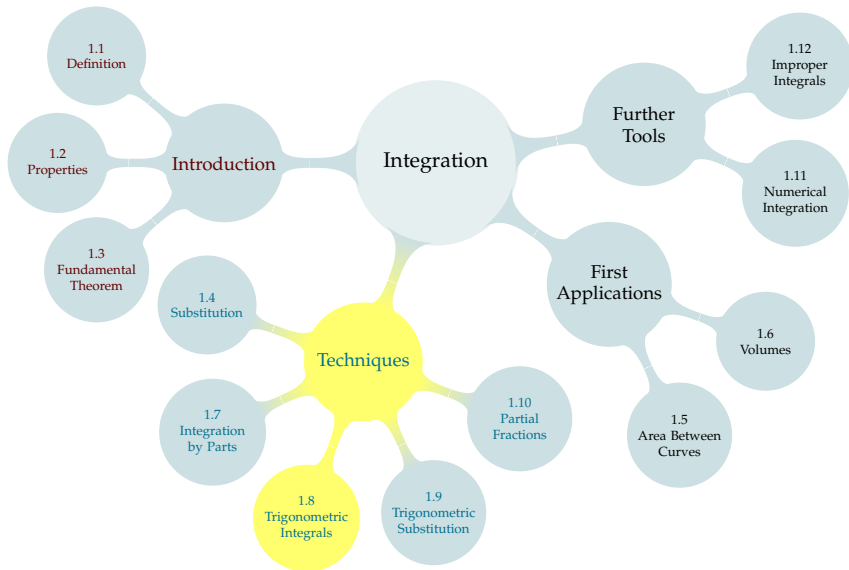


TABLE OF CONTENTS



Evaluating $\int \tan^m x \sec^n x \, dx$

To evaluate $\int \tan^m x \sec^n x \, dx$, we can use:

- ▶ $u = \sec x$ if m is odd and $n \geq 1$
- ▶ $u = \tan x$ if n is even and $n \geq 2$
- ▶ $u = \cos x$ if m is odd
- ▶ $u = \tan x$ if m is even and $n = 0$
(after using $\tan^2 x = \sec^2 x - 1$, maybe several times)

Remaining case: n odd and m is even.

Evaluating $\int \tan^m x \sec^n x \, dx$

To evaluate $\int \tan^m x \sec^n x \, dx$, we can use:

- ▶ $u = \sec x$ if m is odd and $n \geq 1$
- ▶ $u = \tan x$ if n is even and $n \geq 2$
- ▶ $u = \cos x$ if m is odd
- ▶ $u = \tan x$ if m is even and $n = 0$
(after using $\tan^2 x = \sec^2 x - 1$, maybe several times)

Remaining case: n odd and m is even.

The general remaining case is known, but complicated. Instead of treating it exhaustively, we'll show examples of two methods.

$$\int \sec x \, dx$$

We saw a way of integrating secant with the following trick:

$$\begin{aligned}\int \sec x \, dx &= \int \sec x \left(\frac{\sec x + \tan x}{\sec x + \tan x} \right) dx = \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} dx \\ &= \int \frac{1}{u} du \quad \text{with } u = \sec x + \tan x\end{aligned}$$

Another trick: this time let $u = \sin x$, $du = \cos x \, dx$:

$$\int \sec x \, dx$$

We saw a way of integrating secant with the following trick:

$$\begin{aligned}\int \sec x \, dx &= \int \sec x \left(\frac{\sec x + \tan x}{\sec x + \tan x} \right) dx = \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} dx \\ &= \int \frac{1}{u} du \quad \text{with } u = \sec x + \tan x\end{aligned}$$

Another trick: this time let $u = \sin x$, $du = \cos x \, dx$:

$$\begin{aligned}\int \sec x \, dx &= \int \frac{1}{\cos x} dx = \int \frac{\cos x}{\cos^2 x} dx \\ &= \int \frac{1}{1 - \sin^2 x} \cos x \, dx = \int \frac{1}{1 - u^2} du\end{aligned}$$

The integrand $\frac{1}{1-u^2}$ is a rational function of u (i.e. a ratio of two polynomials). There is a procedure, called Partial Fractions, that can be used to evaluate all integrals of rational functions. We will learn it in Section 1.10.



$$\int \sec^3 x \, dx$$

We can integrate around in a circle (with integration by parts) to evaluate $\int \sec^3 x \, dx$. Let $u = \sec x$, $dv = \sec^2 x \, dx$. Then $du = \sec x \tan x \, dx$ and $v = \tan x$.

$$\int \sec^3 x \, dx$$

We can integrate around in a circle (with integration by parts) to evaluate $\int \sec^3 x \, dx$. Let $u = \sec x$, $dv = \sec^2 x \, dx$. Then $du = \sec x \tan x \, dx$ and $v = \tan x$.

$$\begin{aligned}\int \sec^3 x \, dx &= \sec x \tan x - \int \sec x \tan^2 x \, dx \\&= \sec x \tan x - \int \sec x (\sec^2 x - 1) \, dx \\&= \sec x \tan x - \int \sec^3 x \, dx + \int \sec x \, dx \\&= \sec x \tan x - \int \sec^3 x \, dx + \log |\sec x + \tan x| + C'\end{aligned}$$

$$2 \int \sec^3 x \, dx = \sec x \tan x + \log |\sec x + \tan x| + C'$$

$$\int \sec^3 x \, dx = \frac{1}{2} (\sec x \tan x + \log |\sec x + \tan x|) + C$$

with $C = C'/2$.